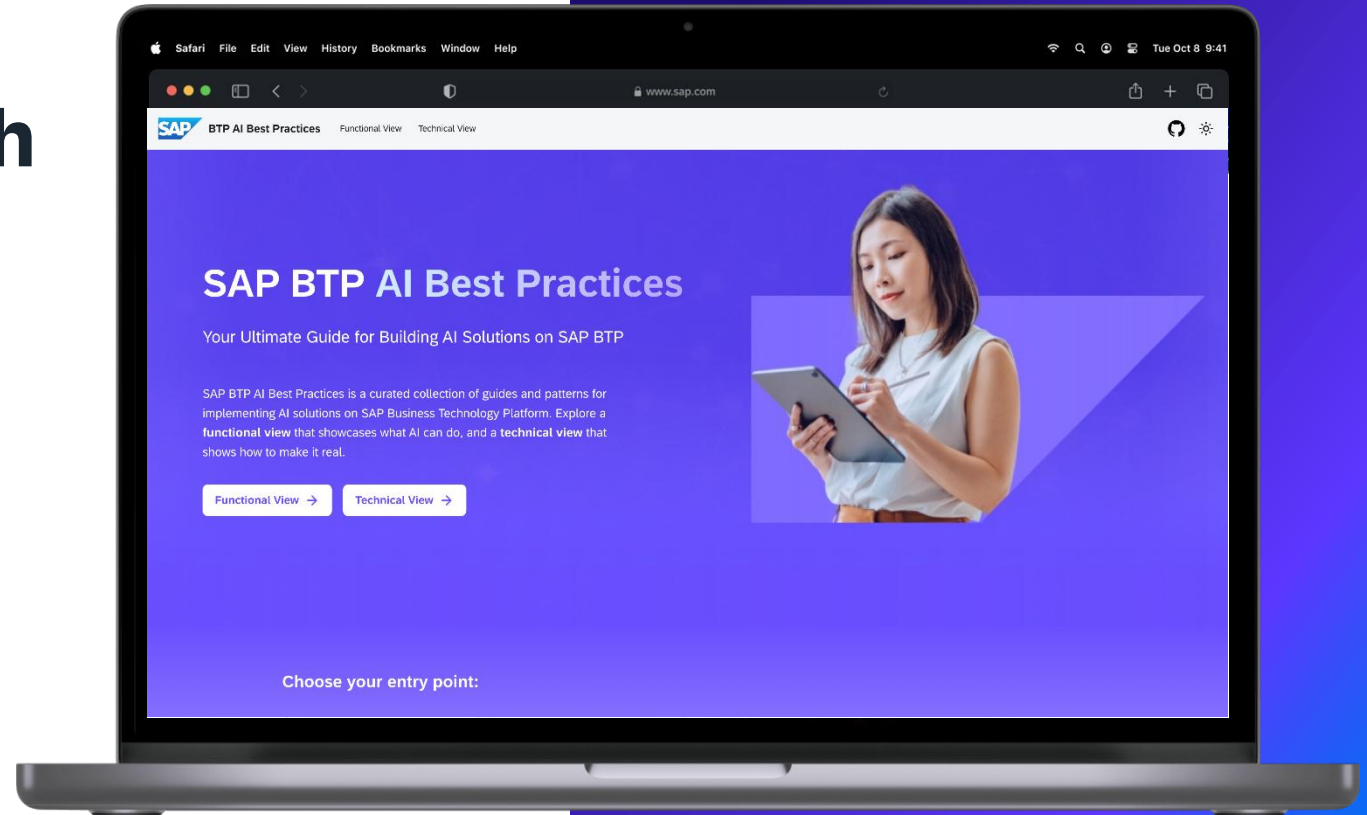


SAP BTP AI Best Practices

Content-based Agents with SAP Joule Studio Agent Builder

A best-practice guide to building grounded, enterprise-ready Joule agents with document grounding, tool orchestration, and SAP-native governance.



BTP AI Services Center of Excellence

29.04.2026

Steps

- 1 Overview**
- 2 Pre-requisites**
- 3 Key Choices and Guidelines**
- 4 Implementation examples**

Why content-based agents?

Ground enterprise answers and actions in approved business content.



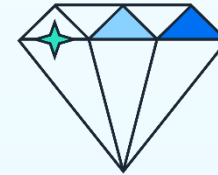
Ground

Use approved documents and business context as the reference for answers



Orchestrate

Combine agent reasoning with Joule skills, MCP servers, and subagents



Govern

Run in the user security context with traceability, approvals, and control

Business Value

Trusted, document-aware responses
Reusable enterprise execution across SAP and non-SAP systems
Security, compliance, and maintainability by design

Expected outcome and benefits

What should a well-designed content-based agent deliver?

Expected Outcome

- Agent definition that Joule can identify for the right intent
- Governed tools and document grounding
- User-context execution with authorization checks
- Testing and debugging before deployment

Benefits

- Organization-specific capabilities
- Reuse of SAP Build artifacts
- Controlled SAP and non-SAP integration
- Managed access and lifecycle

Key concepts

What building blocks matter in Joule Studio?

Joule Studio: capability in SAP Build for creating and managing custom Joule skills and agents.

Joule agent: interactive capability for complex or multi-step tasks.

Agent definition: description, expertise, instructions, and additional context.

Tools: Joule skills, MCP servers, subagents, and document grounding resource groups.

Model Settings: provider and model selection, including an optional backup provider.

Agent Execution Steps: optional pre-processing and post-processing.

SAP Cloud Identity Services and principal propagation: user context and authorization checks end-to-end.

Pre-requisites

Commercial

- SAP Build active user license for developers
- SAP Build subscription with the build-default plan
- SAP Cloud Identity Services subscription
- SAP AI Core and object store, if document grounding is used

Technical

- Setup SAP Business Technology Platform (SAP BTP) subaccount ([Setup Guide](#))
- Set up Joule Studio using the supported setup option ([Guide](#))
- Assign required Joule Studio roles to users ([Guide](#))
- Configure destinations for action projects and connected systems ([Guide](#))
- Configure MCP server destinations, if MCP tools are used ([Guide](#))
- Configure document grounding, if content-based sources are used ([Guide](#))

SAP Business Technology Platform (SAP BTP)

- SAP Business Technology Platform (BTP) is an integrated suite of cloud services, databases, AI, and development tools that enable businesses to build, extend, and integrate SAP and non-SAP applications efficiently.

SAP Build and Joule Studio

- Joule Studio is set up in SAP Build. Initial setup requires the correct active user license for SAP Build and a subscription to the build-default plan

SAP Cloud Identity Services

- Destinations in SAP BTP are used to connect Joule Studio projects and action projects to required backend systems and external services.

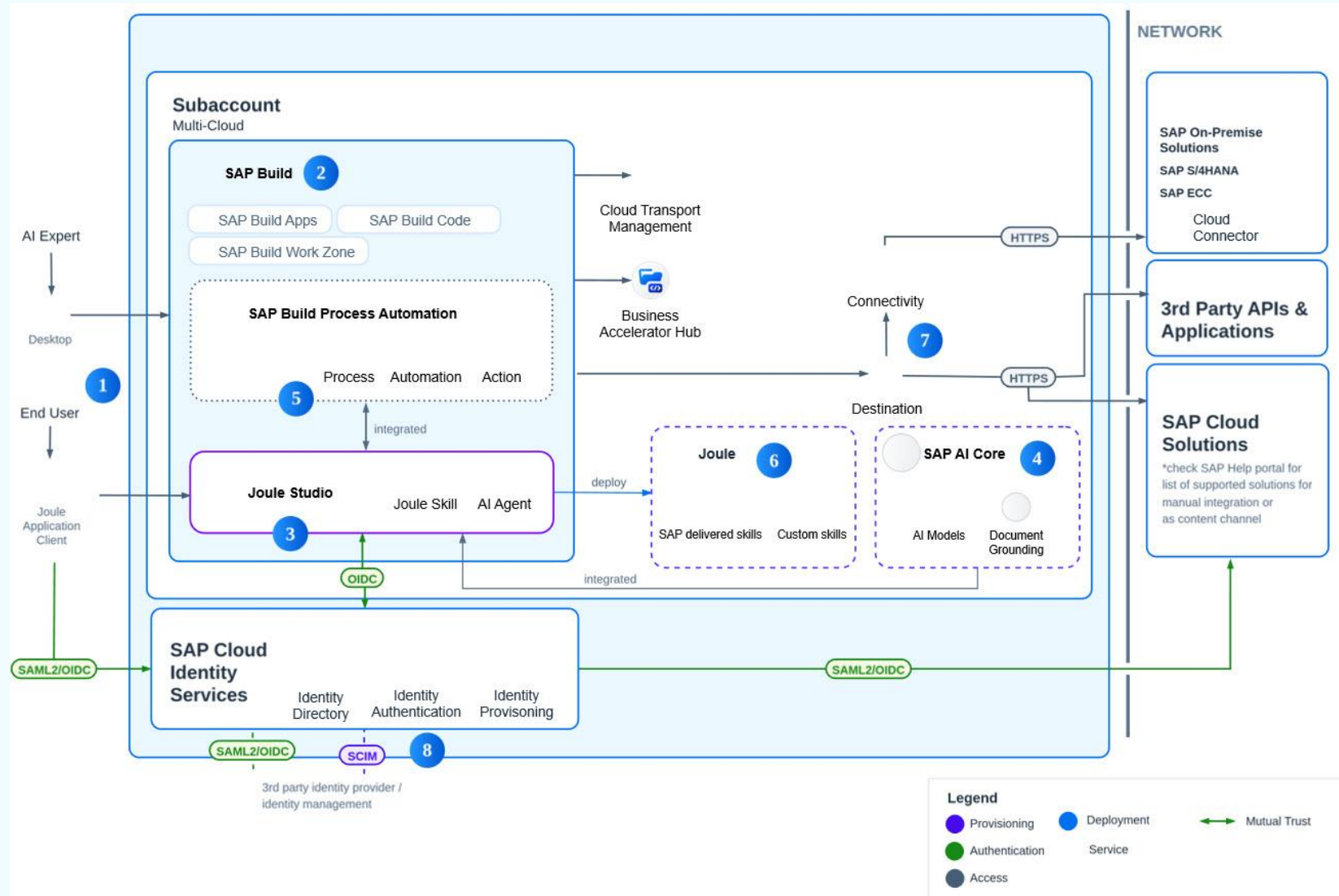
Destinations and connected systems

- Destinations in SAP BTP are used to connect Joule Studio projects and action projects to required backend systems and external services.

SAP AI Core and document grounding

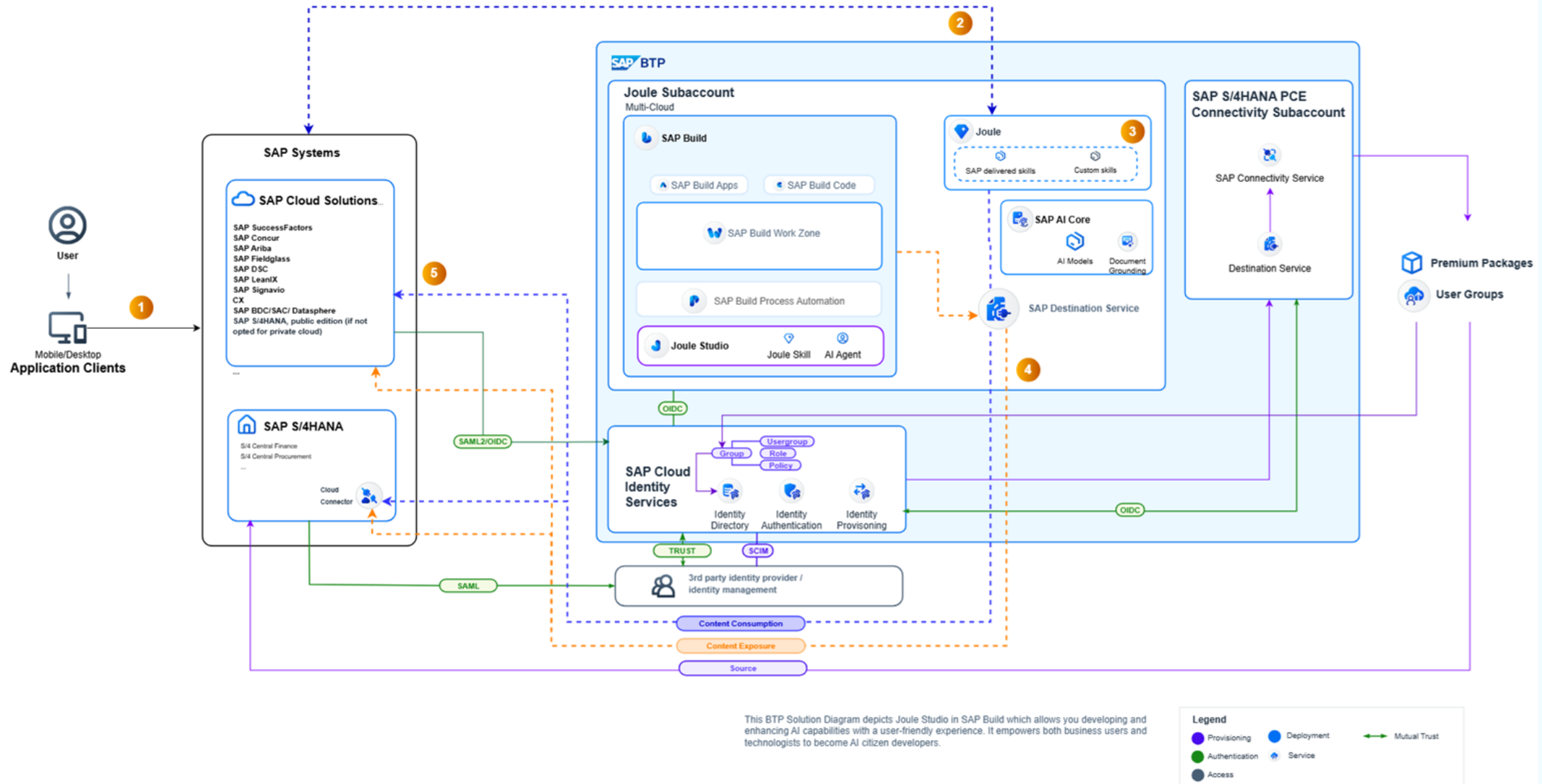
- For content-based agents, document grounding can be configured so agents can use enterprise documents as a source of context. SAP AI Core and an object store are required when document grounding is used.

High-level reference architecture



Extend Joule with Joule Studio

High-level reference architecture



Key Choices and Guidelines

What should you decide before deployment?

1. Scope and orchestration
2. Agent definition
3. Tools and document grounding
4. Security and human-in-the-loop
5. Operations and troubleshooting

Scope

Definition

Tools

Controls

Operations

Scope and orchestration strategy

Clear scope, reusable tools, and Joule Studio orchestration

Definition guidance

- Define the business task, required inputs, and expected outcomes.
- Use agent orchestration when the workflow is better handled by a main agent and specialized agents as tools.
- Use Joule skills for stable, rule-based operations with clear inputs and outputs.
- Keep execution logic in tools and skills; keep the agent instructions focused on coordination and decisions.
- Check Joule Studio constraints for subagents before using orchestration.

Agent definition quality

Description, expertise, instructions, and additional context

Definition guidance

- Use the Description to explain when Joule should activate the agent and which business requests it supports.
- Use the Name as a short, task-oriented identifier.
- Define Expertise as the agent's role and business domain.
- Write Instructions for behavior, tool use, and handling missing information.
- Use Additional Context for examples, SAP terminology, and supplementary response guidance.

Example

Instructions

`Classify the given email and create a ticket using the createTicket tool based on the classification result.`

Recommended tools

`If the question is about X then prefer using the open api tool.`

Final Answer

`When giving your final answer ensure that it includes the status codes of the open-api responses.`

Tooling strategy

Tools like an execution layer

Tooling guidance

- Use Joule skills as tools for well-defined operations with clear inputs and outputs.
- Connect MCP servers when the agent needs tools exposed by external applications.
- Add document grounding as a tool when the agent needs indexed enterprise documents as context.
- Use Model Settings to select the LLM provider, base model, advanced model, and execution steps.
- Keep agent instructions focused on how tools should be selected and used.

Security and Authorization Model

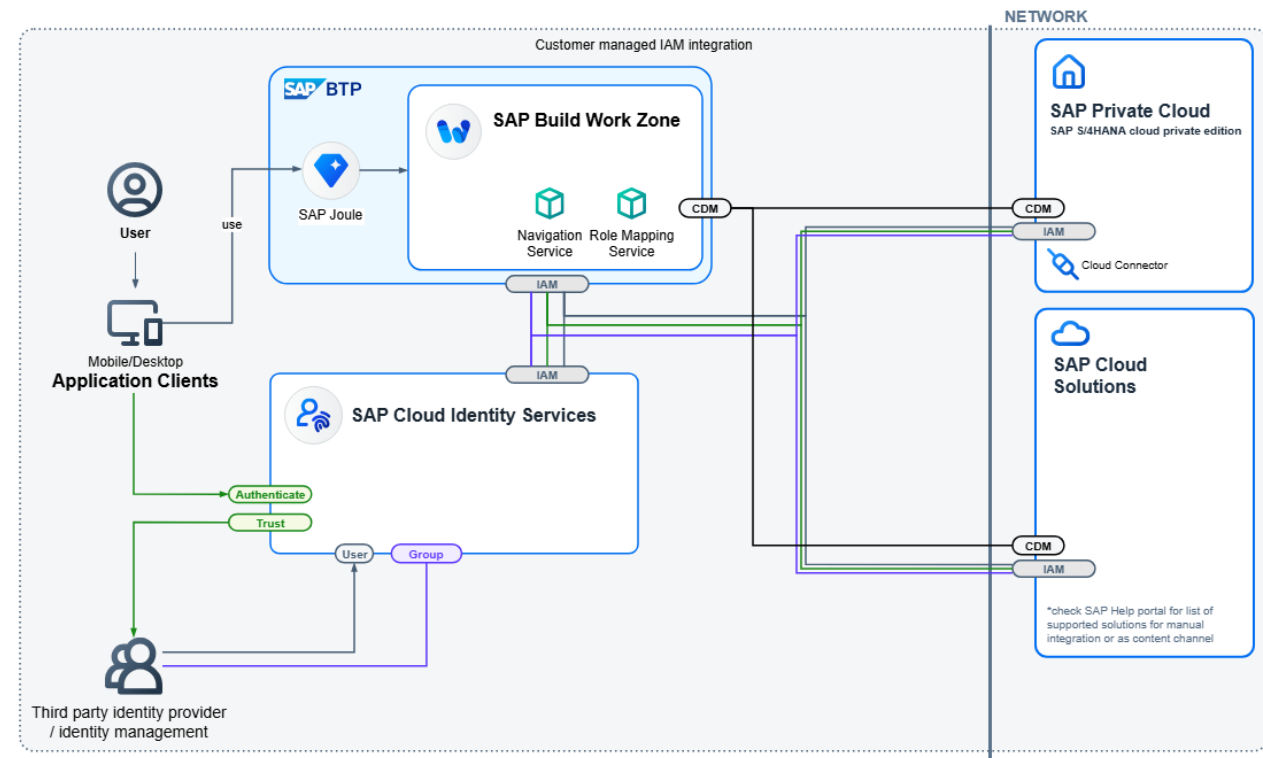
Identity, user context, governed tool access, and audit logging

Security guidance

- Use SAP **Cloud Identity Services** as the identity foundation for Joule and Joule Studio setup.
- Run actions in the user context where principal propagation is used.
- Keep tool and destination access aligned with the agent scope.
- Do not bypass application authorizations or established approval processes.
- Use audit logging and traceable tool execution for security-relevant events.

Identity and Access Management for SAP Joule

SAP IAM integration flows with SAP Joule L0 - Solution Diagram



Recommended usage of SAP Cloud Identity Services for SAP Joule

The diagram shows the major parts of a SAP Joule integration from IAM perspective.

Human-in-the-Loop Controls

Explicit confirmation and governed execution for high-impact actions

Human-in-the-loop guidance

- Define human-in-the-loop interactions for steps that require review, confirmation, or approval.
- Separate recommendations or drafts from execution.
- Ask for explicit confirmation before calling write-capable tools.
- Keep approval steps aligned with the underlying business process.
- Use governed tools and workflows where approvals or manual interventions are required.

Reliability, Testing, and Troubleshooting

Validate inputs, test tool behavior, and use documented diagnostics

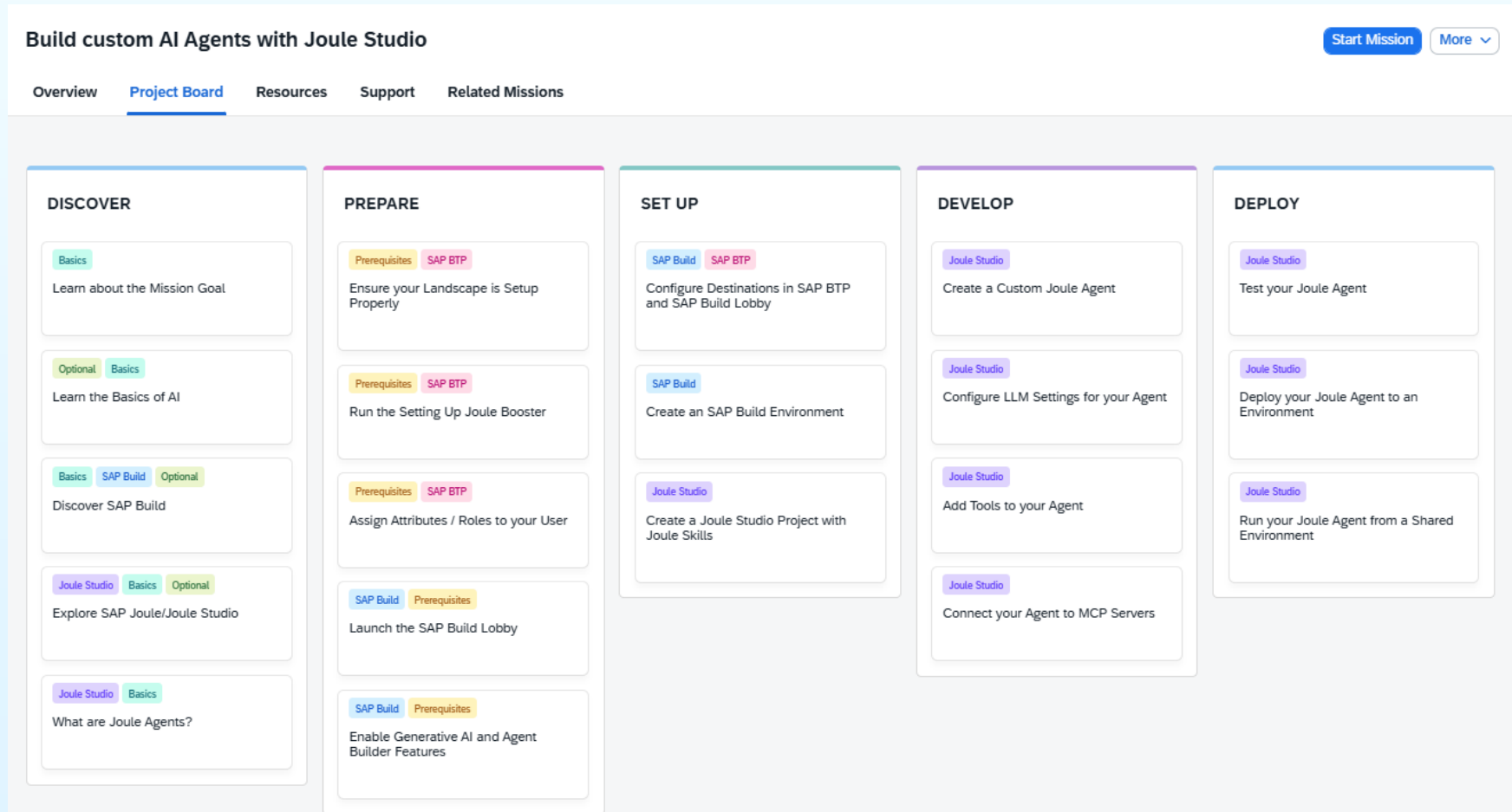
Reliability guidance

- Validate required inputs before invoking tools that read or change business data.
- Start with testing simple parts of the process, then test end-to-end flows.
- Test tool calls, user confirmations, and grounding behavior where they are part of the agent design.
- Check destinations, authorizations, environments, and error messages during troubleshooting.
- Capture the error message and reproduction context before escalation.

Implementation examples

Recommended preparation flow

Discover → Prepare → Set Up → Develop → Deploy



SAP Discovery Center mission flow for building custom AI agents with Joule Studio

Implementation examples: missions and tutorials

Recommended learning path

Recommended

- Get Started with SAP Build Code and Joule using Generative AI (Trial Account) - [link](#)
- Set Up Joule Studio and start with Joule Skills and Agents in BTP Enterprise Account – [link](#)
- Build custom Joule skills for SAP and non-SAP systems using Joule Studio - [link](#)
- Build custom AI Agents with Joule Studio - [link](#)

Additional

- Use Joule Studio with Document Grounding - [link](#)
- Boosting AI-driven Business Transformation with Joule Agents - [link](#)

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Thank you